



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

FACTORING
Practice Problems

1. Factor: $x^2 + 9x + 20$
2. Factor: $2x^2 + 7x + 3$
3. Factor: $3x^2 - 4x - 5$
4. Factor: $x^2 - 49$
5. Factor: $16y^2 - 81$
6. Factor: $x^3 - 27$
7. Factor: $8a^3 + 27$
8. Factor Completely: $6x^2 + 19x + 10$
9. Factor Completely: $4x^2 - 25y^2$
10. Factor Completely: $2x^3 - 18x$



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Step-by-step Solution

1. Factor: $x^2 + 9x + 20$

- **Step 1:** Find two numbers whose:

$$\text{Product} = 20$$

$$\text{Sum} = 9$$

These numbers are **4 and 5**.

- **Step 2:** Write the factors.

$$(x+4)(x+5)$$

Answer: $(x+4)(x+5)$

2. Factor: $2x^2 + 7x + 3$

- **Step 1:** Find two numbers whose:

$$\text{Product} = 2(3) = 6$$

$$\text{Sum} = 7$$

- **Step 2:** Split the middle term (7x).

$$2x^2 + 6x + x + 3$$

- **Step 3:** Factor by grouping.

$$2x(x + 3) + 1(x + 3)$$

Answer: $(2x+1)(x+3)$

3. Factor: $3x^2 - 4x - 5$



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- **Step 1:** Find two numbers whose:

$$\text{Product} = 3(-5) = 15$$

$$\text{Sum} = -4$$

There are no integers that have a product of -15 and a sum of -4. Therefore, this expression $(3x^2 - 4x - 5)$ has no factors and is called a **prime polynomial** or an **irreducible polynomial**.

4. Factor: $x^2 - 49$

- **Step 1:** Identify the squares.

$$x^2 - 49 = x^2 - 7^2$$

- **Step 2:** Use the special product formula (Product of the difference of two squares).

$$(a + b)(a - b) = a^2 - b^2$$

- **Step 3:** Write the factors.

$$(x + 7)(x - 7)$$

Answer : $(x + 7)(x - 7)$

5. Factor: $16y^2 - 81$

- **Step 1:** Identify the squares.

$$16y^2 - 81 = 4y^2 - 9^2$$

- **Step 2:** Use the special product formula (Product of the difference of two squares).

$$(a + b)(a - b) = a^2 - b^2$$

- **Step 3:** Write the factors.

$$(4y + 9)(4y - 9)$$



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Answer : $(4y + 9)(4y - 9)$

6. Factor: $x^3 - 27$

- **Step 1:** Identify the cubes.

$$x^3 - 27 = x^3 - 3^3$$

- **Step 2:** Use the special product formula.

$$(a - b)(a^2 + ab + b^2) = a^3 - b^3$$

- **Step 3:** Substitute.

$$(x - 3)(x^2 + 3x + 9)$$

Answer : $(x - 3)(x^2 + 3x + 9)$

7. Factor: $8a^3 + 27$

- **Step 1:** Identify the cubes.

$$8a^3 + 27 = (2a)^3 + 3^3$$

- **Step 2:** Use the special product formula.

$$(a + b)(a^2 - ab + b^2) = a^3 + b^3$$

- **Step 3:** Substitute.

$$(2a + 3)(2a^2 - 6a + 9)$$

Answer : $(2a + 3)(2a^2 - 6a + 9)$



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8. Factor Completely: $6x^2 + 19x + 10$

- **Step 1:** Find two numbers whose:

$$\text{Product} = 6(10) = 60$$

$$\text{Sum} = 19$$

These numbers are **15 and 4**.

- **Step 2:** Split the middle term (19).

$$2x^2 + 15x + 4x + 10$$

- **Step 3:** Factor by grouping.

$$3x(2x + 5) + 2(2x + 5)$$

Answer: $3x(2x + 5) + 2(2x + 5)$

9. Factor Completely: $4x^2 - 25y^2$

- **Step 1:** Identify the squares.

$$4x^2 - 25y^2 = (2x)^2 - (5y)^2$$

- **Step 2:** Use the special product formula (Product of the difference of two squares).

$$(a + b)(a - b) = a^2 - b^2$$

- **Step 3:** Write the factors.

$$(2x + 5y)(2x - 5y)$$

Answer : $(2x + 5y)(2x - 5y)$

10. Factor Completely: $2x^3 - 18x$



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- **Step 1:** Identify the GCF (Greatest Common Factor).

$$\text{GCF} = 2x.$$

$$\text{Factor out } 2x = 2x(x^2 - 9)$$

- **Step 2:** Use the special product formula (Product of the difference of two squares).

$$(a + b)(a - b) = a^2 - b^2$$

Write the factors.

$$(x^2 - 9) = (x + 3)(x - 3)$$

- **Step 3:** Write the factors.

$$2x(x + 3)(x - 3)$$

Answer: $2x(x + 3)(x - 3)$